Saarthi Manual

Section 2: Components Required + System Design Guide

In this section, we outline every single component required for the Saarthi Smart Stick project, with deep justification, cost estimation, usage purpose, and arrangement plan to tackle the limitations of the WeWALK cane — especially the inability to detect descending stairs and rainy weather challenges.

We solve these problems with:
1. Downward-facing ultrasonic sensors for stair detection.
2. Fully waterproof casing and a two-part architecture (Stick + Waist Pouch) to keep electronics safe.

🛠️ Component List with Cost and Justification:

Component	Use/Justification	Estimated Price (INR)
Raspberry Pi 5	Main processor (Free for you)	₹0
Google Coral USB Accelerator	AI model acceleration with .tflite	₹6000
Night Vision Pi Camera	Real-time input, better in low light	₹900–₹1200
Waterproof Enclosure Box	Safety of electronics	₹250
Speaker Module / Earphones	Audio alerts for user	₹200
Power Bank (10,000 mAh or more)	Power source for Pi + Coral	₹800–₹1200
Ultrasonic Sensors (x3)	Front, downward (stairs), side	₹300 total
Jumper Wires + Soldering Kit	Wiring, permanent fixing	₹400
Foldable Lightweight Stick	Physical stick structure	₹300–₹500
Waist Pouch	Holds electronics, balances weight	₹250
Waterproofing Material (tape, etc)	Rain-proofing all sensors and electronics	₹200

✅ Estimated Total Cost (excluding free Pi): ₹9,000–₹10,000

□ System Architecture Overview

Stick Setup:

- Mount 1 ultrasonic sensor at hand height, facing forward.
- Mount 1 sensor facing downward at a 30–45° angle (for detecting stairs).
- Place speaker module at top near the handle.

Waist Pouch Setup:

- Use waterproof box to hold Raspberry Pi and Coral USB.
- Insert camera facing forward from pouch or belt height.

- Power bank sits beside or below Pi.
- Keep wires tightly packed, covered with waterproofing material.

Connectivity Map:

- Ultrasonic → GPIO pins on Pi
- Camera → CSI port
- Coral USB → USB 3.0 port
- Speaker → 3.5mm or USB Audio adapter

💡 Step-by-Step Assembly Strategy

1. **Sensor Mounting on Stick**
    - Drill or 3D-print mounts for sensors.
    - Use industrial-grade glue to fix ultrasonic sensors.
    - Wrap connections with waterproof tape.
2. **Electronics in Pouch**
    - Insert Raspberry Pi in a plastic enclosure.
    - Use short USB cables for Coral USB and Power bank.
    - Keep cables inside the pouch with minimal movement.
3. **Power Supply**
    - Use power bank with two output ports.
    - One for Pi, another for Coral if needed.
    - Ensure power bank fits well in pouch.
4. **Testing**
    - Write simple Python script for ultrasonic.
    - Test audio output with sample `espeak` or `pyttsx3`.
    - Confirm camera and object detection model works.
5. **Final Waterproofing**
    - Cover stick completely with plastic coating or tape.
    - Put water-resistant bag over pouch if needed.